Deep Learning Based Cervical Spine Bones Detection: A case study using YOLO

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Abstract— **Cervical spine bones detection plays a crucial role in various medical applications, such as diagnosis, surgical planning, and treatment assessment. Traditional methods for cervical spine bones detection often rely on manual identification and segmentation, which are time-consuming and prone to errors. In recent years, deep learning approaches have shown great potential in automating the detection process and achieving high accuracy. In this research paper, we propose a deep learning-based approach for detecting cervical spine bones. Our suggested approach employs the YOLOv5 architecture, a cutting-edge object identification system renowned for its effectiveness and precision. The model is trained to recognize and locate bones structures using computed tomography (CT) scans image of the cervical spine as inputs. We conduct extensive evaluations using the trained models on the cervical spine dataset. The mean average precision (mAP) scores achieved by our model are 93% at threshold (mAP_0.5) and 83% at thresholds ranging from (mAP_0.5:0.95), which demonstrate the effectiveness of our approach in accurately detecting and localizing cervical spine bones. Our deep learning-based method for detecting cervical spine bones with high mAP scores presented in this research paper has significant implications for medical applications. With accurate and reliable bones detection, medical professionals can enhance diagnosis, surgical planning, and treatment assessment processes. The achieved mAP scores showcase the performance and potential of our proposed method, contributing to the advancement of bone detection techniques in cervical spine imaging and facilitating collaboration between the medical imaging and deep learning communities.**

Keywords— **Cervical Spine, Bones Detection, Computer Vision, Deep Learning, Computed Tomography (CT), YOLOv5**

I. Introduction

The field of medical imaging has experienced a dramatic revolution in recent years as a result of significant breakthroughs in deep learning and computer vision technology. Among the many applications of these breakthroughs, the recognition and interpretation of anatomical features inside medical images has received a lot of attention [1]. The detection of cervical spine vertebrae is critical in the diagnosis and treatment of spinal diseases and injuries [2]. The cervical spine, which consists of seven vertebrae labelled C1 to C7, is crucial in supporting the skull, protecting the spinal cord, and allowing for a wide variety of head motions [3]. The accurate and efficient identification of these vertebrae is critical for the examination of disorders such as fractures, degenerative disc diseases, and congenital anomalies [4]. Traditionally, radiologists were responsible for manually identifying cervical vertebrae in radiography pictures, a time-consuming and error-prone operation that might be vulnerable to inter-observer variability. To overcome these issues, deep learning-based techniques for automated object detection have emerged as viable alternatives. In this regard, the You Only Look Once (YOLO) family of object detection models, specifically YOLOv5, has attracted attention due to its real-time capabilities and improved accuracy in object localization tasks [5]. Because of its ability to detect many objects at the same time with a single network pass, it is a good option for automating the recognition of cervical spine vertebrae in CT scan images.

This study investigates the use of YOLOv5 in medical imaging, specifically the detection of cervical spine vertebrae in CT scan images. We intend to use deep learning to create a robust and efficient system that can assist radiologists in detecting anomalies or injuries in the cervical spine with high accuracy and in a short amount of time. We aim to contribute to the ongoing efforts to enhance patient care and outcomes in the fields of radiology and spinal health by fusing the cutting-edge technology of deep learning with the wealth of information present in CT scan images.

II. Materials and Methods

A. Dataset Description

In our study, we used the RSNA 2022 Cervical Spine Fracture Detection Challenge dataset [6]. This comprehensive dataset is made up of 2019 CT scans that are specifically focused on the cervical spine area. The availability of segmentations for a portion of the scans is an intriguing feature of the dataset. The provided segmentation labels range from 1 to 7, representing the seven cervical vertebrae from C1 to C7.

B. Data Pre-processing

We used the segmentation data found in the dataset to help us find the cervical spine bone. This segmentation distinguished the different cervical vertebrae, giving each one a specific numerical label that ranged from 1 to 7, as well as a background label that was specified as 0. The next step was to draw bounding boxes. Using the segmentation information, we precisely delineated bounding boxes around each cervical vertebra. These bounding boxes effectively enclosed the areas of interest inside the images. Most importantly, we labeled the slices according to the cervical spine number for a total of 9170 images. Based on the segmentation information, each image was paired with the numerical label of the cervical vertebra to which it referred. This technique endowed each image with a clear and meaningful label, allowing the YOLOv5 model to be trained for precise recognition of cervical spine bone.

C. YOLOv5 Network

YOLO (You Only Look Once) has gained fame as a prominent deep neural model for real-time object detection [7], owed to its robust performance in swiftly identifying objects within a real-time environment. The original YOLO version was introduced by Joseph Redmon and his research team in 2016. It distinguishes itself as a single neural network architecture capable of not only detecting object bounding boxes but also predicting the probability of object classes within an image. Unlike earlier object detection tools such as CNN, RCNN, and Faster-RCN [8, 9], YOLO offers a unique advantage in terms of both speed and accuracy, outperforming even mask-RCNN, which excels in precise object detection but lags in processing speed [10]. YOLO is particularly well-suited for scenarios that demand swift and accurate object detection, such as in military applications where speed and precision are critical. It's a single-network solution known for its robustness and accuracy. Over time, YOLO has seen several iterations, including YOLOv2, YOLOv3, v4, and v5, each enhanced with new features and functionalities [11, 12, 13]. YOLOv5 was released in 2020, and it is fast and highly accurate, with various network architectures and data augmentation techniques to enhance performance [14,15]. The YOLO network architecture is made up of three key parts: the backbone, the neck, and the head. The backbone includes modules such as Focus, Convolution with batch normalization and Leaky ReLU (CBL), Mix Conv (Mix convolution), Cross Stage Partial Network (CSP), and Spatial pyramid pooling (SPP). It receives input images with a resolution of 512 x 512 x 3 via the Focus structure, applies slicing operations to minimize image size, and performs convolutional operations with kernels. The backbone outputs are used as input for the next phase of the YOLOV5 design, the neck network. The Neck Network is made up of CBL, CSP, Concat, and Up-sampling. It produces three outputs, which are used as inputs to the head, also known as the detector. This layer makes predictions by displaying the bounding box around the target items, classification of the objects, and probability of the object belonging to a given class.

III. Experiments

This section presents the experimental details of this research study. The constructed model was able to detect the targeted in the images, namely cervical spine vertebrae. The model accurately labelled the objects and categorized the vertebrae by drawing bounding boxes around them.

A. Training YOLOv5

The Yolov5 versions were trained in this study, with YOLOv5l being the fastest among them. The dataset used for training consisted of 9170 images, with 80% of the data allocated for model training. A batch size of 16 and an image size of 512 x 512 were chosen. The training utilized a learning rate of 0.01, with a learning rate decay of 0.0005, and the SGD optimizer was employed for network optimization. Training was performed using the train.py script, specifying parameters such as epochs, batch size, and model weights. To facilitate model training, specific folder structures were required. The dataset folders were divided into two, one containing image and the other containing labels in text file format. These label files provided information about object classes, bounding boxes, coordinates, height, and width of the bounding boxes, with one class of objects per line. The experimental parameters are detailed in Table 1 below.

TABLE 1. EXPERIMENTAL PARAMETERS

Parameters	Values
Epochs	200
Image size	512
Learning rate	0.01
Batch size	16
Optimizer	SGD
Momentum	0.937
Weight decay	0.0005

B. Result and Discussion

The study used all four versions of YOLOv5, namely YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. The findings of a comparative analysis revealed that YOLOv5x performed better than the other models. Values of 0.869, 0.906, 0.933, and 0.839 were obtained for the precision, recall, mAP_0.5, and mAP_0.95 measures, respectively.

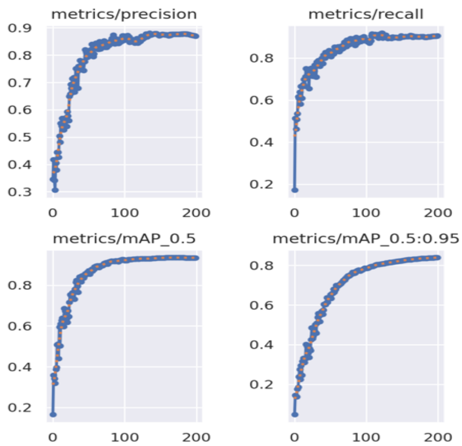

Figure 1. Precision, recall, mAP_0.5, and mAP_0.5:0.95 of YOLOv5x

In Figure 1, the precision value exhibited a gradual increase from 0.346 to reach 0.869, while the recall initially started at 0.171, experienced a sudden rise, and then stabilized with increasing epochs, ultimately reaching 0.906. Additionally,

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Figure 1 also illustrates mAP @0.5 and mAP @0.5:0.95, which were measured at 0.933 and 0.839, respectively.

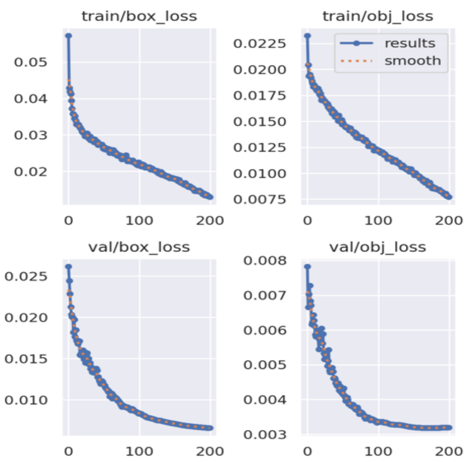

Figure 2. Box loss and object loss for the training and validation, respectively.

In Figure 2, both box loss and object loss are depicted for both training and validation phases. During training, the box loss steadily decreases with increasing epochs, starting at 0.057 at epoch 0 and reaching 0.012 by epoch 200. Similarly, object loss decreases from 0.023 to 0.0075 as the model reaches 200 epochs. In contrast, during validation, the box loss decreases slightly from 0.026 to 0.06 as the model reaches 200 epochs, showing minimal improvement. Concurrently, object loss starts at 0.0028 during validation and gradually decreases with increasing epochs, reaching 0.0027 by epoch 200.

We employed YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x models and conducted training using the cervical spine dataset. The performance of these models was assessed using various evaluation metrics, and the outcomes are presented in Table 2. This table offers a comparative analysis of different YOLO architectures, showcasing the results through the evaluation of parameters such as precision and recall.

TABLE 2. COMPARISON OF YOLO MODELS

Model	**Precision**	**Recall**	**mAP_0.5**	**mAP_0.95**
YOLOv5s	0.825	0.857	0.892	0.715
YOLOv5m	0.843	0.909	0.921	0.794
YOLOv5l	0.862	0.887	0.919	0.803
YOLOv5x	0.869	0.906	0.933	0.839

Performance metrics including mAP, precision, and recall were computed for all four YOLO architectures, and the experimental findings revealed that YOLOv5x outperformed other YOLO versions. Specifically, YOLOv5xl achieved an mAP_0.5 of 93.3 and an mAP_0.95 of 83.9, highlighting its superior performance.

Figure 3 provides a visual representation of the detection outcomes achieved on the validation dataset, and the remarkable level of accuracy depicted in the results serves as a strong indicator of the exceptional performance exhibited by our model.

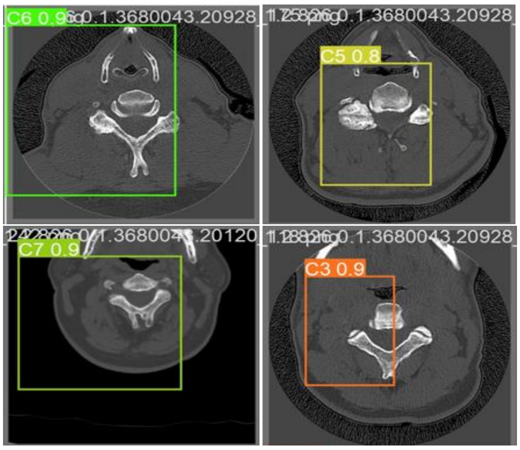

Figure 3. Detection of cervical spine vertebrae

Figure 3 depicts our model's ability to detect cervical bones, specifically C3, C5, C6 and C7. The higher level of accuracy displayed in these detections demonstrates the model's ability to properly recognize these anatomical features within the cervical spine. This level of precision is critical in medical applications where perfect localization of individual vertebrae is required for diagnosis and therapy planning.

Figure 4 displays the confusion matrices of YOLOv5x on our validation dataset. A confusion matrix is a table that is used to evaluate the performance of a classification model. It shows the number of correct and incorrect predictions made by the model compared to the actual outcomes. The matrix is made up of four components: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). These confusion matrices provide a comprehensive overview of the YOLOv5 model's performance by showing the alignment between actual classes and predicted classes.

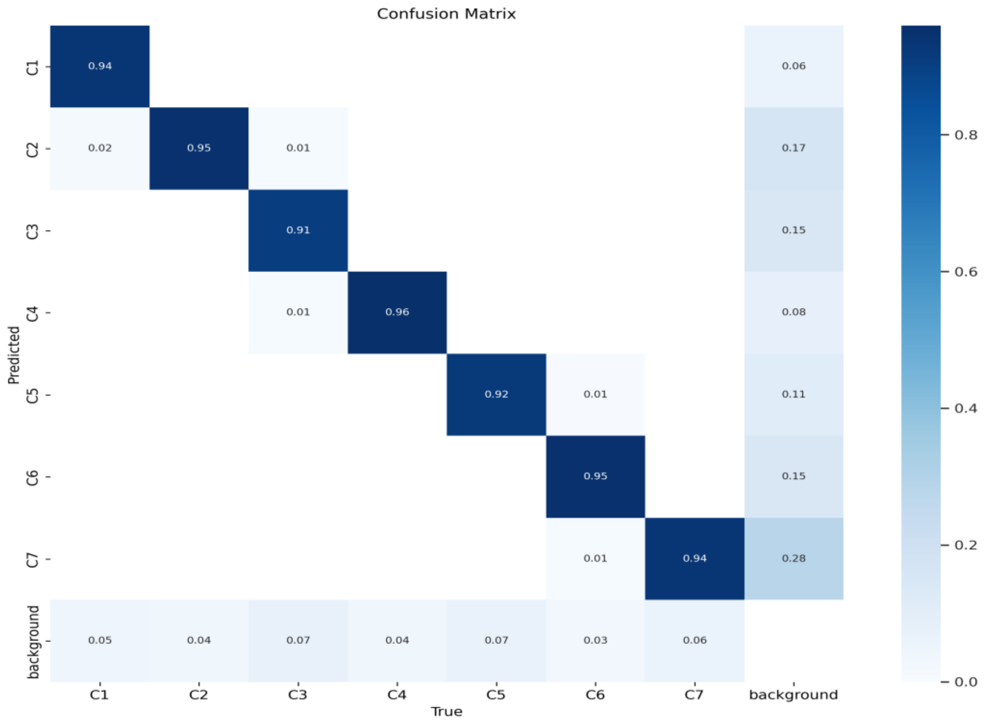

Figure 4. Confusion matrix of YOLOv5x

IV. Conclusions

In this study, using real-time object detection models based on deep learning, we explored the use of YOLO algorithms for the detection of cervical spine structures. The YOLOv5 model, which uses the Pytorch framework and convolutional neural networks as its foundation for feature extraction and object detection, stood out among the other YOLO versions as a highly effective object detection model. We used four distinct YOLOv5 architectures, with YOLOv5x performing particularly well, to identify the cervical spine bones. This model exhibits the capacity to confidently and properly identify vertebrae, which can be a huge help to medical experts when making diagnoses and formulating treatment plans.

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